Chapter 2

AI Subfields

2.1 INTRODUCTION

In **Chapter 1**, we have been introduced to the basic concepts of Artificial Intelligence. In this chapter, we will concentrate on different constituents of AI and a brief outline about each one of them. The grand aim or the ultimate vision of AI has been to simulate human thinking. Before moving forward, it is also essential to differentiate between intelligent thinking and intelligent action. The below table gives a very high-level difference between the two.

Intelligent Thinking	Intelligent Action
Intelligent thinking, also known as cognitive intelligence, refers to the ability of AI systems to simulate human thought processes, reasoning, and problem-solving. **Examples:** Understanding natural language, recognizing patterns, making inferences, and solving complex problems	Intelligent action, also known as behavioural intelligence, pertains to the ability of AI systems to interact with and manipulate the physical world. **Examples:** Decision-making, planning and execution of actions

Hence, a more precise goal of AI is to develop systems which can exhibit human-level or superhuman-level performance in a wide range of cognitive and physical tasks. A more tangible goal can also be to build an intelligent robot.

What are some of the capabilities that such a system will need?

1. Capability to make sense of visual cues (for example, reading the traffic signs in case of a self-driving car). The subfield dedicated to the above is called **computer vision**.
2. Capability to understand natural language and respond in natural language (one of the tasks that is routinely used here is creating a sum-

mary of a given text). Formally such a task is called text summarization. The subfield of AI, that deals with the same is called **natural language processing.**

3. Capability of self-learning (When we want to learn anything new, we often look at a worked-out example and derive some rules in our mind). These examples are often called training data as per computational parlance. The subfield that concentrates on the same is **Machine Learning**. A recent advancement, which deserves mention in its own right is called deep learning. This particular discipline is based on a very fascinating set of frameworks named Deep Neural Networks. One other technique which is very important is reinforcement learning.
4. Capability of capturing the knowledge in some way may be in the forms of some rules. Hence, when we consult a doctor, he has some concrete knowledge about symptoms, diseases and treatments. These construct a knowledge base. Logical Inferences may be made using the knowledge base. Some of the logical formalisms are **predicate logic, propositional logic, first-order logic,** etc. There is another very interesting dimension here. It turns out that in the real world, some information is not measurable but coveys sense, like this is very hot today or this vegetable is quite cheap so on and so Fuzzy Logic, Rough Set deals with the same.
5. Capability of acting on the basis of perceived inputs. Essentially, it is not enough to process a traffic signal showing the speed limit, the break needs to be pressed to slow down. The subfield that deals with the same is called robotics.

At a high level, these capabilities translate to the important subfields of AI. These subfields often intersect and collaborate, and advancements in one subfield often benefit others. AI is a rapidly evolving field, and new subfields and interdisciplinary areas continue to emerge as research progresses.

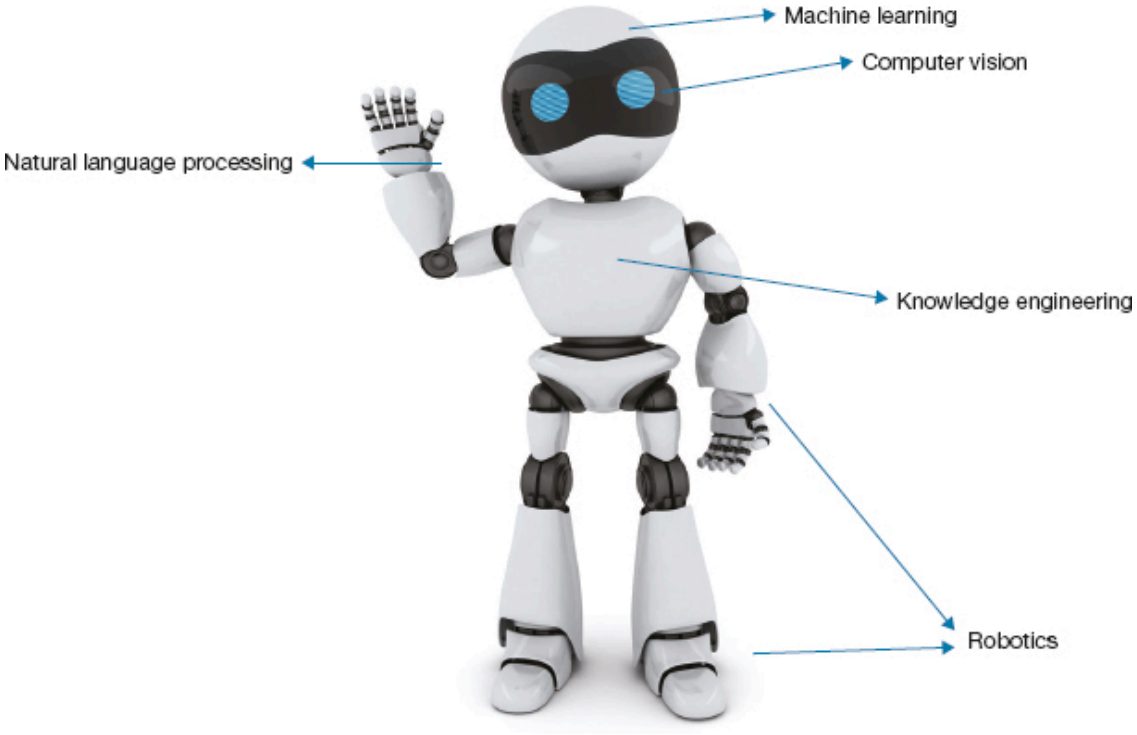


Figure 2.1 Different sub-fields with respect to an artificial man

A tabular summary of the subfields is enclosed below.

Subfield	Description	Applications
Computer vision	Extracts meaning from images and videos.	Object detection, facial recognition, scene understanding
Natural language processing	Deals with the interaction between computers and human languages.	Text classification, machine translation, sentiment analysis
Machine learning	Develops algorithms that can learn from data. Deep learning is a subfield of machine learning.	Image recognition, natural language processing, fraud detection
Knowledge engineering	Involves representation of knowledge and application of logic for inference.	Expert systems
Robotics	Develops robots that can interact with the physical world.	Navigation, manipulation, object recognition

There are many associated issues like explainability of the decisions, fairness and ethical considerations of the AI systems, which will be covered in subsequent chapters.



DO YOU KNOW ?

INDIAai, the National AI Portal of India, is a government-run website launched in May 2020 to provide a central hub for all things related to AI in India. It offers a variety of resources, including news and articles, information on key AI players, insights into the global and Indian AI landscape, resources for students, entrepreneurs, professionals and executives, and a database of AI initiatives and startups in India. INDIAai also plays a key role in promoting AI research and innovation in India by supporting cutting-edge research and applied activities on AI-related priorities.

2.2 COMPUTER VISION

Computer vision is a field of artificial intelligence that gives computers the ability to see, interpret and make sense of the world around them. However, fascinating this sounds it was a herculean task to capture, store, process and finally make sense of the images. Thanks to phenomenal development in storage, processors as well as algorithms, it is a rapidly growing field with a wide range of applications which is discussed hereafter.

2.2.1 Object Detection

When humans look at a photograph or watch a video, they have the innate ability to readily spot people, objects, scenes and visual details. The goal of object recognition is to teach a computer to do what comes naturally to humans: to gain a level of understanding of what an image contains. For example, a computer vision system could be used to detect cars in a traffic scene or people in a crowd.

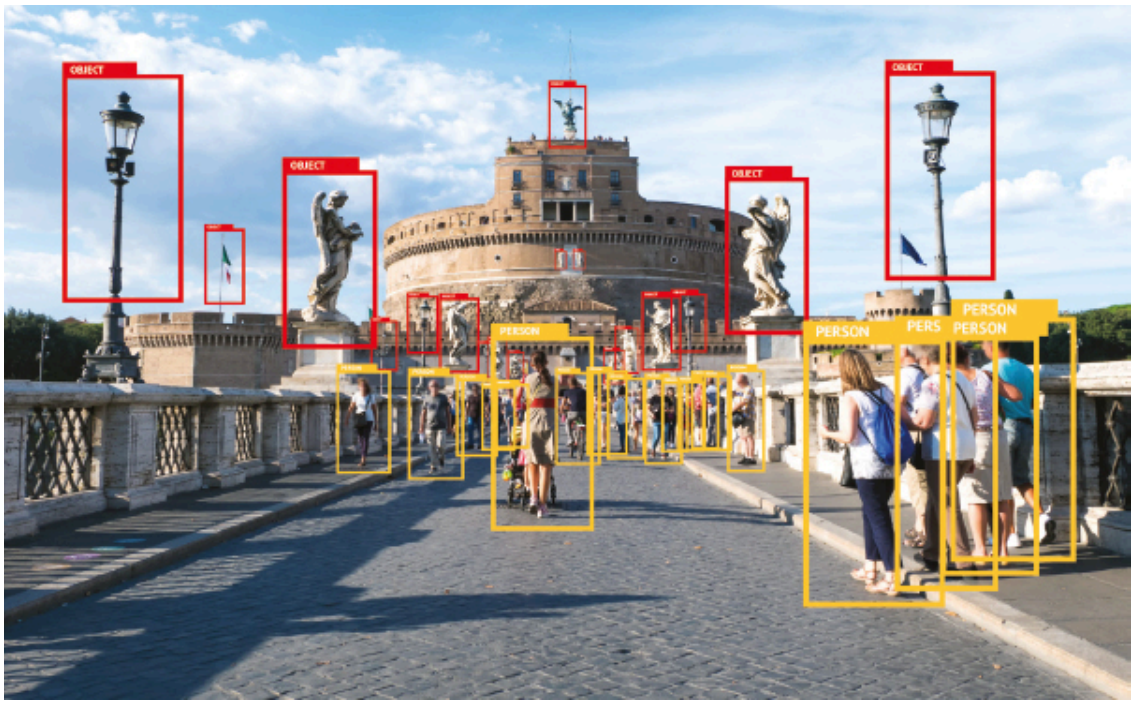


Figure 2.2 Object Recognition

Typically, object recognition consists of two main steps. The first step is called object localization, where a rectangle is put around the objects as shown in **Figure 2.2**. Such a rectangle is known as the bounding box. The second step is object classification. Bounding Boxes for object recognition in an office environment is also shown in the above figure.

Hence, step 1, concentrates on ascertaining whether there is an object in the scene, if yes what type of object it is. Following are some of the examples where object recognition is used across the globe:

- The Mumbai Traffic Police is using object recognition to detect vehicles and pedestrians in traffic. This is done by using cameras to scan the roads and identify objects that are not supposed to be there. If a vehicle or pedestrian is detected, the traffic police may be alerted for taking action to prevent an accident. They have been one of the early adaptors of technologies by installing a large number of CCTV Cameras.
- AI-driven surveillance robots are currently under development by nations such as South Korea and Israel, with the objective of deploying them to oversee border fence security. Here object recognition is being applied for human detection.

- Drishti, a real-time object recognition application developed by researchers and scientists at the Indian Institute of Technology Bombay, serves as a valuable tool for assisting individuals with visual impairments in navigating their surroundings. This innovative application utilizes the smartphone's camera to identify objects within the user's environment, subsequently delivering spoken notifications to describe these objects. For instance, the app can inform users about their proximity to a road crossing, the presence of individuals in their path, or the existence of nearby staircases. Although Drishti remains in the development phase, it has already made a positive impact, with over 10,000 individuals in India having utilized it. Users have lauded the application for its accuracy and user-friendly interface.

2.2.2 Facial Recognition

This is the ability to identify a person's face from an image or video. Facial recognition systems are often used for security applications, such as access control and fraud detection. This is easier said than done, as a person may appear in front of a camera in various attires (with a cap, a sunglass, mask and what not), different backgrounds and varying luminosity. Below are two important implementations of facial recognition.

The Aadhaar system, managed by the Unique Identification Authority of India (UIDAI), is a biometric identification program that includes facial recognition as one of its authentication methods. Aadhaar cards, which contain unique identification numbers, are linked to individuals' biometric data, including facial images, for identity verification in various government and private sector services.

DigiYatra is an initiative by the Ministry of Civil Aviation in India aimed at creating a seamless travel experience for air passengers. It includes a facial recognition-based system for biometric authentication at airports. Passengers who voluntarily enrol in the program can go through various checkpoints, such as check-in, security and boarding gates, using facial recognition instead of traditional identification documents.

Hyderabad Police has more than 7,00,000 Cameras installed in the city. They are using these cameras, transmitters and finally a data centre to

identify known criminals in the streets of the city. The system is going through a lot more developments as of now. Of course, Aadhaar is being used in authentication using face recognition.

Before we move on to other important areas of computer vision, let us clearly differentiate two terms namely face detection and face recognition. The following table summarizes the key differences.

Characteristic	Face detection	Face recognition
Purpose	To identify the presence of a human face in an image or video.	To identify the identity of a person based on their face.
Difficulty	More fundamental and general.	More advanced and specific.
Example	Identifying the faces of your friends in a photo.	Identifying the names of your friends in a photo.

2.2.3 Scene Understanding

This is the ability to understand the context of an image or video. For example, a computer vision system could be used to understand the layout of a room or the activities that are taking place in a scene.

Scene understanding is a more complex task, then the above two. Essentially, if a picture is shown, the textual description of the image will be required for understanding the scene. For example, there may be a scene where Virat Kohli is hitting Stuart Broad for a six, over long on. As you see this will involve detecting the persons, identifying them, detecting other objects, detecting and identifying the location and the activity.

There are a lot of applications of scene understanding. However, the most prominent of them are self-driving cars and AR/VR.

- **Self-driving cars:** Self-driving cars need to have the capability to understand the scene around them in order to make safe driving decisions.
- **Virtual reality and augmented reality:** Virtual reality and augmented reality applications need to be able to understand the scene around the user in order to create a realistic and immersive experience.

2.2.4 Medical Imaging

Computer vision is used extensively in medical imaging to analyse images of the human body. For example, systems/algorithms based on computer vision can be used to detect cancer cells in images of tissue or to diagnose diseases from X-rays. There are several imaging techniques like X-Rays, Computed Tomography (CT), Magnetic resonance imaging (MRI), Ultrasound, Positron emission tomography (PET), etc., which are routinely performed to help with diagnosis and effective treatment. AI techniques can be applied to assist doctors, this can also be helpful in a country like India where medical practitioners are in short supply. The initial interpretation can be done by an AI-based medical diagnosis system and then only priority cases can be routed to medical professionals.

For interested readers, there is a dedicated society the Medical Image Computing and Computer Assisted Intervention Society (MICCAI - **http://www.miccai.org/**). It also organizes a very reputed conference with the same name.

The pandemic has created a lot of interest in computer vision-based medical imaging techniques. Some of the benefits offered by computer vision based covid-19 detection are as follows:

- **Improved accuracy:** Computer vision can help to improve the accuracy of diagnosis by identifying patterns in images that are difficult for humans to see.
- **Reduced costs:** Computer vision can help to reduce the costs of diagnosis by automating tasks and freeing up radiologists to focus on more complex tasks.
- **Improved patient care:** Computer vision can help to improve patient care by providing more accurate diagnoses and personalized treatment plans.

All new technologies will also have some challenges associated. Enlisted below are some of these challenges.

- **Data availability:** Computer vision algorithms require large datasets of labelled images to train. However, these datasets can be difficult

and expensive to obtain.

- **Accuracy:** Computer vision algorithms are still not as accurate as human radiologists. However, the accuracy of computer vision algorithms is improving all the time.
- **Interpretability:** It can be difficult to interpret the results of computer vision algorithms. This can make it difficult for radiologists to trust the results of computer vision algorithms.

Computer vision is a complex field, but it is also one of the most promising areas of artificial intelligence. As computer vision technology matures, we can expect to see even more amazing and important applications in the future.

2.2.5 Prominent Applications of Computer Vision

Self-driving Cars

Computer vision is used to help self-driving cars navigate the road and avoid obstacles. Many companies are working on self-driving car research. Alphabet which is the parent company of Google has been working with Waymo (**Fig. 2.3**). Tesla is another important player in this segment. App cab company Uber has also made significant investments in this. Some of the concerns regarding self-driving cars are cost of the cars, safety aspects as well as regulatory aspects of driving these cars.



Figure 2.3 WAYMO self-driving car

Virtual Reality/Augmented Reality (VR/AR)

Computer vision is used to create realistic virtual worlds that can be explored by users. Virtual reality creates a completely immersive experience by blocking out the real world and replacing it with a computer-generated environment. Users wear a headset that displays the virtual environment, and they can interact with it using controllers or their own bodies. Augmented reality overlays computer-generated images onto the real world. Users see the real world through a device like a smartphone or headset and they can interact with the computer-generated images by touching them or moving their heads.

An even more exciting concept is **metaverse** which goes beyond the realm of Individual AR/VRs, here multiple people can socialize and participate in the activity in a digital universe. While AR/VR serves as the building blocks, the metaverse needs many more technologies like block chain, cloud computing, etc.

2.3 NATURAL LANGUAGE PROCESSING

Natural language processing (NLP) is a subfield of artificial intelligence that deals with the interaction between computers and human (natural)

using languages. More simplistically, this subfield of AI focuses on developing the capability of machines to interpret natural language like a human being. Before we discuss some of the important applications. We will later on see, that the algorithms expect some kind of disciplined data. Data that has a structure, i.e., has a definite set of columns and rows with clear meaning. Text may be very informative, but it lacks structure. Text pre-processing comes to our rescue here.

Text pre-processing is an essential step in natural language processing (NLP) that involves cleaning and preparing text data for analysis and modelling. Some common pre-processing techniques in text include:

- **Lowercasing:** Converting all text to lowercase to ensure consistency and prevent case-related duplication (e.g., 'word' and 'Word').
- **Tokenization:** Splitting text into individual words or tokens, makes it easier to analyse.
- **Stop Word Removal:** Eliminating common, uninformative words (e.g., 'and,' 'the,' 'is') that don't contribute significantly to the analysis.
- **Punctuation Removal:** Stripping text of punctuation marks, which are often irrelevant for many NLP tasks.
- **Numerical and Special Character Removal:** Removing numbers and other non-alphabetic characters, depending on the analysis.
- **Whitespace Trimming:** Removing unnecessary spaces, tabs, or line breaks.
- **Lemmatization and Stemming:** Reducing words to their base or root form to standardize and consolidate related words (e.g., 'running' to 'run').
- **Spell Checking and Correction:** Identifying and rectifying spelling errors in the text.
- **Handling Contractions and Abbreviations:** Expanding contractions (e.g., 'can't' to 'cannot') and standardizing abbreviations.
- **Handling HTML Tags:** Stripping or converting HTML tags in text data.
- **Text Normalization:** Normalizing text data to ensure consistency, for example, converting different date formats to a common standard.
- **Removing or Masking Personal Identifiable Information (PII):** Replacing or removing sensitive information like names, addresses, or social security numbers for privacy and compliance.

- **Removing URLs and Email Addresses:** Eliminating URLs and email addresses that might not be relevant for analysis.
- **Text Segmentation:** Splitting text into segments or paragraphs, depending on the analysis requirements.
- **Sentence and Document Length Normalization:** Ensuring that sentences or documents are of consistent lengths for tasks like text classification.
- **Encoding and Decoding:** Converting text between different character encodings as necessary.

Some of the important tasks in the domain of natural language processing are as follows.

2.3.1 Text Classification

Text classification involves categorizing or assigning predefined labels to a piece of text. For example, sentiment analysis, where text is classified as positive, negative, or neutral based on the expressed sentiment, or topic classification, where text is categorized into specific topics or domains. A typical sentiment analysis AI model will take a piece of text as input and categorize them as positive, negative or neutral. Well, this is for a simplistic understanding, there can be of course many variants and complexities. Following are a couple of examples of sentiment analysis:

- Very good, solid, good balance, comfortable...loved it - > Positive
- Parts are missing and I have given this as a gift and now return window is closed. Very bad experience - > Negative

Topic classification can be used to categorize a news or an article to a given topic, which can help to organize the topics.

2.3.2 Named Entity Recognition (NER)

NER focuses on identifying and extracting specific named entities from text, such as names of persons, organizations, places, dates, or other predefined entities. It helps in information extraction and understanding the context of a text. When a machine is reading a piece of text, NER can go a long way in understanding the text.

Some examples of NER are as follows:

1. 'The President of the United States is Joe Biden.' The NER system would identify 'President' as a title, 'United States' as a location and 'Joe Biden' as a person.
2. 'The company TCS was founded in 1968.' The NER system would identify 'TCS' as an organization and '1968' as a date.
3. '**The meeting is going to be held at 10**:00 AM today.' The NER system would identify '10:00 AM' as a time and 'today' as a date.

NER is a critical component of many NLP systems, such as chatbots, sentiment analysis tools and search engines. It is also used in a variety of other applications, such as:

- **Information extraction:** NER can be used to extract information from text, such as the names of people, organizations, and locations. This information can then be used for a variety of purposes, such as building a knowledge base or generating reports.
- **Machine translation:** It can be used to improve the accuracy of machine translation by identifying named entities in the source text and translating them correctly.
- **Text summarization:** NER can be used to improve the accuracy of text summarization by identifying named entities in the text and summarizing them correctly.

2.3.3 Question Answering

Question Answering (QA) systems aim to automatically answer questions posed in natural language. These systems analyse questions, search for relevant information and generate appropriate answers. QA systems can be based on retrieving answers from a database or using a machine comprehension approach to understand and generate answers.

There are two main types of QA tasks:

1. **Extractive QA:** In extractive QA, the system extracts the answer to the question directly from the text. For example, if the question is 'What is

the capital of France?', the system would extract the answer 'Paris' from the text.

2. **Abstractive QA:** In abstractive QA, the system generates an answer to the question that is not directly extracted from the text. For example, if the question is 'What is the meaning of life?', the system might generate an answer that is a philosophical or religious reflection on the meaning of life

Based on the expertise of the QA System, there can be another set of classifications.

- **Closed-domain QA:** These systems focus on answering questions within a specific domain, such as medical QA or legal QA.
- **Open-domain QA:** These systems aim to answer questions on a wide range of topics without a specific domain restriction.
- **Knowledge base QA:** Some QA systems are built on top of structured knowledge bases like DBpedia or Freebase, allowing them to answer fact-based questions more effectively.

QA systems are deployed in a variety of applications, including the following:

- **Search engines:** QA systems can be used to answer questions that users ask in search queries.
- **Virtual assistants:** QA systems can be used to answer questions that users ask virtual assistants, such as Amazon Alexa or Google Assistant.
- **Knowledge bases:** QA systems can be used to answer questions that users ask about a knowledge base, such as a database of product information or a database of medical information.

QA systems can very naturally be used in Quora. It might be of interest to the readers that a Kaggle Competition (**https://www.kaggle.com/c/quora-question-pairs**) was launched around 2017. Here, the task that was given to solve was identifying question pairs which are similar. In fact, the latest favourite of everyone, ChatGPT is also an example of a QA System.

2.3.4 Machine Translation

Machine translation focuses on automatically translating text or speech from one language to another. It involves understanding the source language, generating an intermediate representation and producing the translated output in the target language. The machine translation is done mainly following three approaches:

1. **Rule-based MT:** Rule-based MT systems use a set of rules to translate text. The rules are typically based on the grammar of the two languages and the context of the words in the text.
2. **Statistical MT:** Statistical MT systems use statistical methods to translate text. The systems learn to translate text by analysing a large dataset of parallel text.
3. **Neural MT/AI-based MT:** Neural MT/AI Based MT systems use artificial neural networks to translate text. Artificial neural networks are a type of machine learning model that has a special ability to learn complex patterns in data.

There is a lot of focus from the Central Government on machine translation from one Indian language to another. Interested readers may visit **www.bhasini.gov.in** for more details. The objective is digital empowerment and digital inclusion by making the ecosystem available in most of the Indian languages. One notable initiative in this regard is named Bhasha Daan. Here, all citizens can contribute to building an open-source dataset.

Google Translate has been there for long time now and is supporting more than 120+ languages worldwide.



Figure 2.4 Machine Translation using Google Translator

Well, the above example (**Fig. 2.4**) is at a toy scale. Google Translator is doing that translation for 24 × 7 over numerous examples and are collecting active feedback and improving itself further. But genuinely language is a very complex thing and the current state of the art is indeed good but far from perfect.

2.3.5 Text Generation

Text generation involves creating coherent and meaningful text in natural language. This can range from simple tasks like generating product descriptions to more complex applications like story generation or language modelling. Similar, to other NLP tasks, the primary three approaches of text generation are (i) rule based methods (ii) statistical methods and (iii) neural or AI-based methods.

Text generation has a variety of use cases. Couple of crucial ones are mentioned below:

- **Chatbots:** Chatbots are AI-based programs that can simulate conversations with humans. Chatbots are often employed in customer service applications, where they can answer questions and provide support to customers.
- **Content creation:** Text generation capability can be leveraged to create content, such as articles, blog posts and social media posts. This can be useful for businesses that want to create content that is engaging and informative. Shown in **Fig. 2.5**, is a conversation with ChatGPT, where the transformer-based language engine generates po-

etry like a professional.

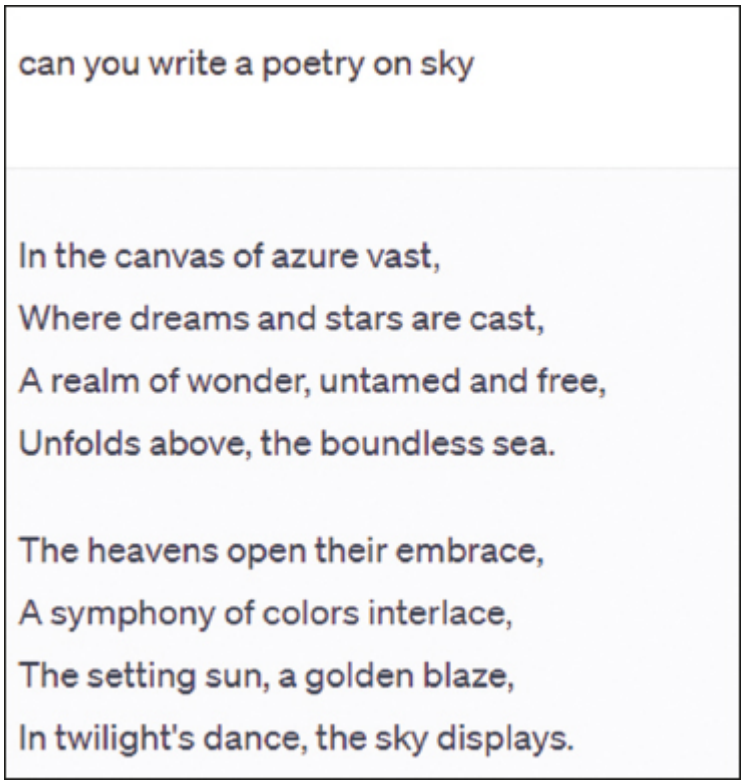


Figure 2.5 Content Generation Example using a Conversational Agent

2.3.6 Text Summarization

Text summarization aims to generate concise summaries of longer texts, condensing the main points or key information. Summarization techniques can be extractive (selecting and rearranging important sentences) or abstractive (generating new sentences to capture the essence of the text).

There are two main approaches to text summarization which is discussed hereunder.

Extractive Summarization

Extractive summarization involves selecting and extracting sentences or phrases directly from the original text to form the summary. The extraction is typically based on various criteria, such as sentence importance, relevance to the main topic and coherence with other selected sentences. The advantage of extractive summarization is that it maintains the exact wording and context from the original text, but it can sometimes result in less fluency and coherence in the summary.

Abstractive Summarization

Abstractive summarization, on the other hand, involves generating new sentences that may not appear in the original text to form the summary. In this approach, the model understands the content of the input text and rephrases or rewrites the information in a concise and coherent manner to create a summary. Abstractive summarization can produce more fluent and human-like summaries, but it is generally more challenging due to the need for language generation and maintaining the intended meaning.

Text summarization methods can use various techniques, including statistical approaches, rule-based methods, and machine learning, similar to the case of machine translation.

Applications of text summarization include:

- **News summarization:** Automatically generating concise summaries of news articles to provide users with quick and informative overviews.
- **Document summarization:** Condensing lengthy documents, research papers, or reports into shorter summaries for easy understanding and reference.
- **Social media summarization:** Summarizing long social media posts or discussions to highlight the main points or sentiments.
- **Content curation:** Summarizing multiple pieces of content to provide users with curated and relevant information.

Text summarization plays a crucial role in helping users navigate through vast amounts of textual data efficiently and can significantly save time and effort in information retrieval and understanding.

2.4 MACHINE LEARNING

Machine learning is a sub-discipline of artificial intelligence, which depends heavily on data. In a typical scenario, when we write a computer program for let's say finding whether a number is prime or not, we give an input, an integer in this case to the computer program and the pro-

gram gives an output either true or false. So, if an input 7 is given to the program, the output will be true and for any composite number let's say 20, the output will be false. The logic of checking primality needs to be coded in some programming language. The basic logic may be something like this, for the number we are testing, we will try to divide it by all numbers between 1 and the number itself. If it is divisible by anyone of them, then it is not a prime number. The logic we wrote in natural language just now, can be written more concisely. Such a representation of the logic, which is human readable is called an algorithm, which is often the first step of writing a program.

One other thing, that needs to be discussed in this context is that machines do not understand human languages, they can only understand strings of 0s and 1s. These are called binary strings. However, humans cannot express logic in terms of 0s and 1s. There are several programming languages which has their own grammar. If it is close to machine language then it is a low-level language and if it is close to natural language then it is a high-level language.

A program like the above-mentioned can easily be written in a high-level language. Let's look at the problem statement a little more analytically. Here the objective of the program is clear and the steps to achieve the same are also clear. There are several toy problems like this:

1. Swapping the value of two variables
2. Find maximum of three numbers
3. Finding the factorial of a given integer
4. Find the greatest common divisor of two numbers
5. Find whether a particular year is a leap year or not

The programming paradigm that we discussed works fine for all five problems. However, let's look at some other types of problems.

1. A customer is approaching a bank for a loan, looking at his demographic information, and past financial record, can we say whether he is going to default or not.
2. Classifying an email as a normal email or spam email.
3. Predicting whether tomorrow there will be rainfall or not

There is a clear difference here. The problem definition is clear, but the exact steps to find the same is not clear. These are the problems that are attempted to be solved using **machine learning.** Typically, this is learned from past examples. For the first problem, past customers data will be looked at. **Table 2.1** enlists some sample customers.

Table 2.1 Loan Default Sample Data

CId	Gender	Age	Marital Status	Monthly Salary	Loan Default
1	M	30	Single	30000	Y
2	F	41	Married	40000	N
3	F	31	Married	25000	N
4	M	27	Single	60000	Y
5	M	52	Married	70000	N

Gender, age, marital status and monthly salary are called input features or attributes. Loan default is called the output attribute or target attribute, also called the label. A machine learning program looks at the data and learns patterns on its own to associate the input with the output. These past examples serve as the training set for the machine learning program, which now develops a capability, given the attributes of a customer it can predict whether the customer will default or not. Such type of learning through examples is called inductive learning.

If we attempt to categorize, the machine learning tasks, it will look like **Fig. 2.6**.

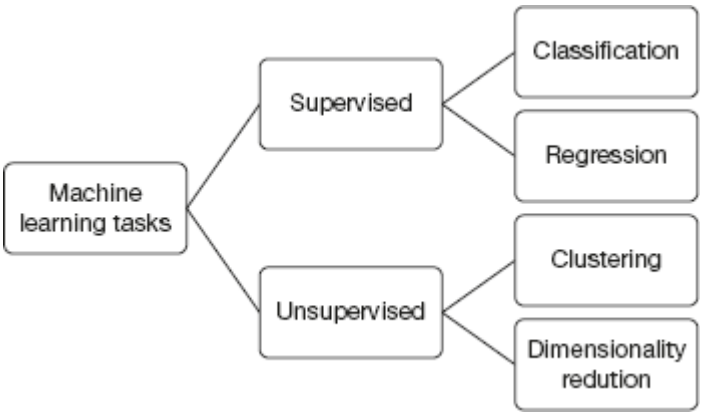


Figure 2.6 Categories of Machine Learning Tasks

This is a very simplistic representation. Many more categories can be added as a refinement to the above classification. However, it serves as a baseline.

2.4.1 Classification

Classification is the task of categorizing input data into predefined classes or categories. The model is trained on labelled data, where each data point is associated with a specific class label.

Examples of classification tasks include spam detection, sentiment analysis, image classification and text categorization.

Figure 2.7 shows a sample dataset of playing golf.

	Outlook	Temp	Humidity	Windy	Play Golf
2	Rainy	Hot	High	FALSE	No
3	Rainy	Hot	High	TRUE	No
4	Overcast	Hot	High	FALSE	Yes
5	Sunny	Mild	High	FALSE	Yes
6	Sunny	Cool	Normal	FALSE	Yes
7	Sunny	Cool	Normal	TRUE	No
8	Overcast	Cool	Normal	TRUE	Yes

Figure 2.7 Golf Dataset

This is a popular playing golf dataset. Outlook, temperature, humidity and windiness are the input attributes and playing golf is the target attribute. So, the task to be done is to develop a model, based on the given data about a particular day, whether it will be good for playing golf. There are

several classification algorithms. One such algorithm is also called decision tree. This is shown in **Figure 2.8**.

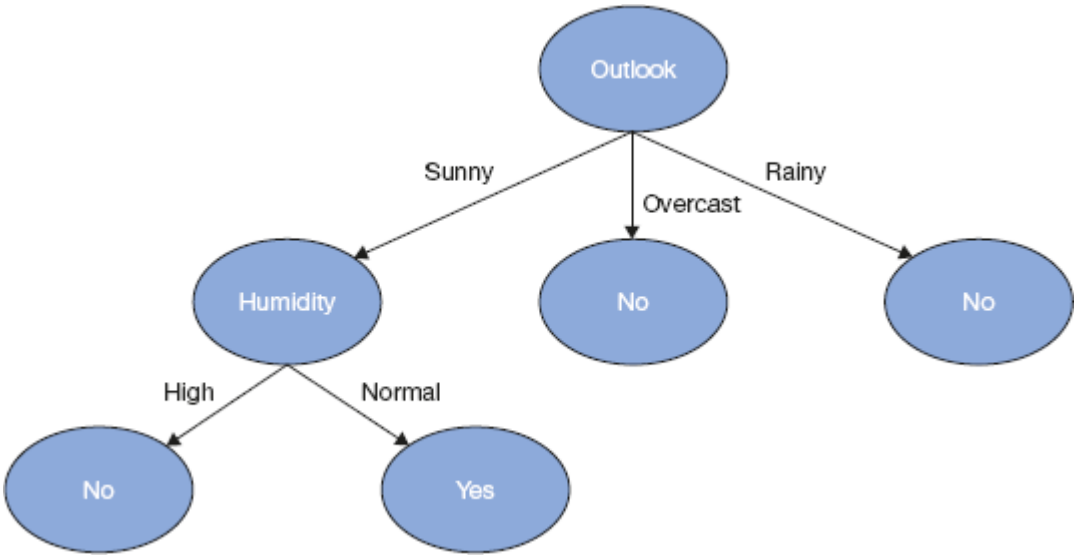


Figure 2.8 Partially Constructed Decision Tree on Golf Dataset

So, if on a particular day, the outlook is sunny and the humidity is normal, irrespective of other attributes, the decision is to play (YES). You might observe that the above representation does not look like a tree, well it's an inverted tree. The node on the top is called the root node and the decision nodes i.e., the last level of nodes are called leaf nodes.

These decision trees are used in two phases. In phase 1, they learn from previous data and in phase 2, they can predict. Let's take some other examples. Let's say, we are working with a problem of predicting whether a particular employee will stay in the organization or leave the organization. So, in phase 1, a decision tree will be constructed from past data (previous employee records their demographic and behavioural data and a target field, whether they stayed or left), and then in phase 2, when a new employee's record is given, using the decision tree, it can be predicted whether the employee will be staying or leaving.

Hence, the decision tree can work as a classification model. This model is called a white box model as the reason for reaching the final decision is amply clear. As we move to a more complex model, this becomes opaque and such models are called black box models.

There are several other models for classification namely k nearest neighbour (kNN), Naïve Bayes (based on conditional probability), logistic regression, support vector machine, etc. There is also a neural network-based model named as multi-layer perceptron, which we will discuss later.

2.4.2 Regression

Regression is the task of predicting continuous numerical values based on input data. The model learns the relationship between the input features and the target numerical value during training. Regression is commonly used for tasks such as predicting house prices, sales forecasting and time series analysis. The easiest variation of regression is when we have one input variable and one output variable. The input variable is often called the regressor or predictor. The simplest of the models is called a linear regression. Let's say, we are trying to find out that for a science graduate, based on their CGPA, what should be their in-hand salary. Like, the classification problem, there will be past data about the students.

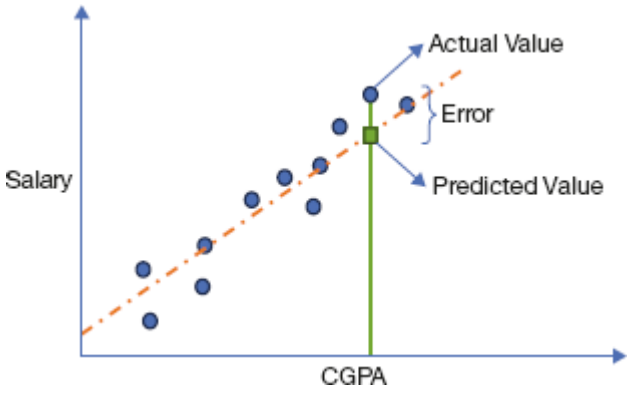


Figure 2.9 Scatter Plot and Regression line Salary vs CGPA

In **Fig. 2.9**, CPGA, the input variable is often called the independent variable is drawn along the X-axis and Salary the dependent variable is drawn along the Y-axis. Each circle above represents data about a past student. Based on these circles, a line has been drawn such that **collectively the circles are as close as possible to the line.** The previous sentence is worth a lot of attention. We can draw several alternative lines, which are close to the circles, however, we are interested about the line, which is closest to all. Some curious readers may want to quantify closeness.

Well, if you look thoroughly for one of the circles at the extreme right, a straight-line projection is shown from the circle to the regression line. The value on the line is the predicted value and the circle is the actual value. Error is given as Actual Value – Predicted Value. When, there is only one input variable it is called simple linear regression and if there are more than one input variable, we call this approach multiple linear regression. These are linear models because the relationship between the dependent and independent is assumed to be linear (the model is a straight line). Instead of a line, it can be a more complex one like a polynomial.

It might be noted that both classification and regression are examples of supervised learning. Who gives the supervision? The past experience embodied in terms of labelled data gives supervision.

2.4.3 Clustering

Clustering is an unsupervised learning task where the goal is to group similar data points together based on their similarity. Data points within the same cluster are more similar to each other than to those data points in other clusters. Clustering is a more difficult task than supervised learning as there is no past data with labels. Hence, the algorithms have to learn on their own.

Figure 2.10 illustrates the dilemma. If someone asks us to group the building block pieces, they can be grouped either by shape or by size or even by colour. Clustering algorithms can be classified further. One way of classification is exclusive and nonexclusive clustering methods. In an exclusive algorithm, a single point can belong to only one cluster and in case of non-exclusive algorithm, a particular point or observation can go to multiple clusters. For example, if one cluster stands for aquatic animals and another cluster stands for landliving animals. A turtle or a frog will belong to both clusters.



Figure 2.10 Kid's building Blocks

One important clustering algorithm is called *k*-means. When applied to a dataset this algorithm can find *k*-clusters. The detailed working is beyond the scope of the current book.

2.4.4 Dimensionality Reduction

Dimensionality reduction is the process of reducing the number of features or variables in the data while preserving its important structure. This helps in visualizing and understanding high-dimensional data and can improve the performance of machine learning models.

Imagine you have a bunch of data, like a list of people's heights, weights, and ages. Now, think of this data as points in a three-dimensional space where each axis represents one of these measurements. It's like a 3D graph where each point represents a person. Dimensionality reduction techniques are a way to look at this data and figure out what's most important. Imagine you want to simplify this 3D graph into a 2D graph but keep as much information as possible. The motivation for dimensionality reduction techniques is that.

Principal Component Analysis (PCA) and t-distributed Stochastic Neighbour Embedding (t-SNE) are popular techniques used for dimensionality reduction. There are many benefits of dimensionality reduction. Firstly, with reduced dimensions, there is less requirement for storage.

Then, the machine learning models that are being built also require less time to be constructed and finally, with a smaller number of features, it becomes much more explainable.

2.4.5 Recommendation Systems

As we live our lives, we consume, use or do a variety of things. That may be home appliances like TV, Fridge, Washing Machine, etc., that can be buying groceries, maybe planning a trip, going for a family dinner, reading a book, watching a movie or applying for a job. You just name it. For all these actions, there are a variety of choices available and that's where the recommender system comes in. It keeps track of your usage, your choices and gives you a recommendation. Maybe you order Chinese food on weekends, so based on that a recommendation comes from the food delivery app, when a new Chinese eatery has opened up and gives a lot of discounts.



DO YOU KNOW

There is an interesting story of how this all started. Netflix (one of the largest subscriptions based streaming service provider), in the year 2006, announced a one Mn $ prize money to any ML researcher, group of researchers who can better their recommendation. This kind of money generated a lot of interest and actually many algorithms got invented for the same. The competition lasted for three long years.

More formally, recommendation systems aim to predict and suggest items or content that a user might be interested in. Two principal categories of recommendation are (a) Content based recommendation (b) Collaborative Recommendation or Filtering.

Content-based recommendation is based on the static characteristics of the item. Taking the inspiration from Netflix, continuing with the selection of movie examples. Content-based recommendation will compare two movies based on their short description, genre, director and actor information etc.

Collaborative filtering is more on the basis of the dynamic characteristics of an entity. Let's say, you like action movies and love thrillers and do not enjoy the romcoms. This technique will quickly identify users who are similar to you and if this group of users thinks highly of a movie (given a high rating) and this is not yet watched by you, it will be recommended to you. This type of filtering which is based on similar types of users is called user-based collaborative filtering.

There is another type. Let's say, you liked movie 1, so similar movies will be found out. How this similarity will be determined? If the same set of users give the same kind of rating to two movies, then they are similar, let's say movie 1 and movie 2 are two such movies. Now, if movie 2 is not watched by you, then it will be recommended to you. Such a type of collaborative filtering is called item-based filtering.

There are some interesting problems as far as recommender systems are concerned. Let's discuss a couple of them here.

Long Tail Problem

In any digital marketplace, there are a lot of merchandise. The count of such products may easily run into thousands. While some of the items are popular and rated by a lot of people, some items are consumed and rated less.

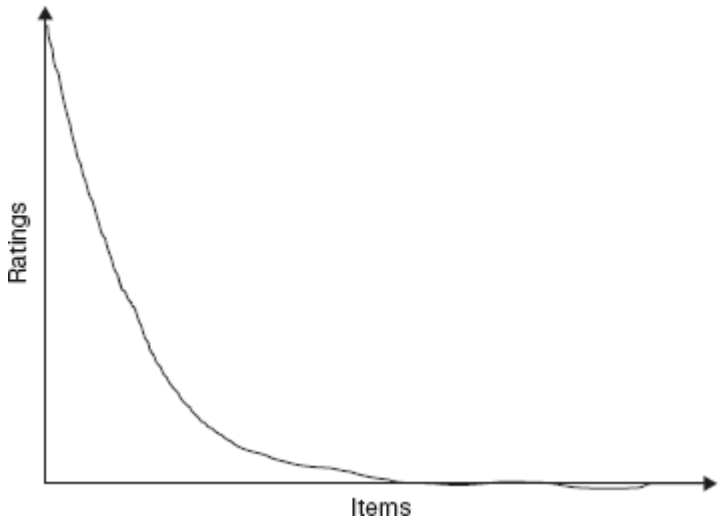


Figure 2.11 Long Tail Distribution of Ratings

The items which are less rated appear at the tail of the distribution and this tail is quite long. The Long tail distribution of product ratings is shown in **Figure 2.11**. Recommending these items becomes quite difficult.

Cold Start

This is a problem when a new item is introduced. There are no ratings available for this item. Hence, recommending such an item also becomes difficult.

2.4.6 Anomaly Detection

Anomaly detection involves identifying unusual or abnormal data points that deviate significantly from the normal pattern. It is used in fraud detection, fault detection and cybersecurity. This is also referred to as outlier detection, novelty detection etc. Some important use cases of anomaly detection are fraud detection, intrusion detection, fake news detection, etc.

Let's further elaborate on this with an example. Generally, every person has a regular way of doing his or her spending. For example for Arnab, it may be ATM withdrawals on Monday, Grocery shopping on weekends, Movies or fine dining maybe once a month, some medical expenses, fees for school, etc. Any transaction which is high value and does not fit in this regular pattern (anomaly) may be thought of as suspicious or fraud.

2.4.7 Deep Learning as a Special Case of Machine Learning

As mentioned earlier, the grand aim of AI is to create an artificial human brain. The basic unit of the human brain is a biological. Hence, the starting step of AI will be to create artificial neurons. Late, 1950's the perceptron model was popularized, which had the capability of simple decision-making. **Figure 2.12** shows a biological neuron with its artificial counterpart.

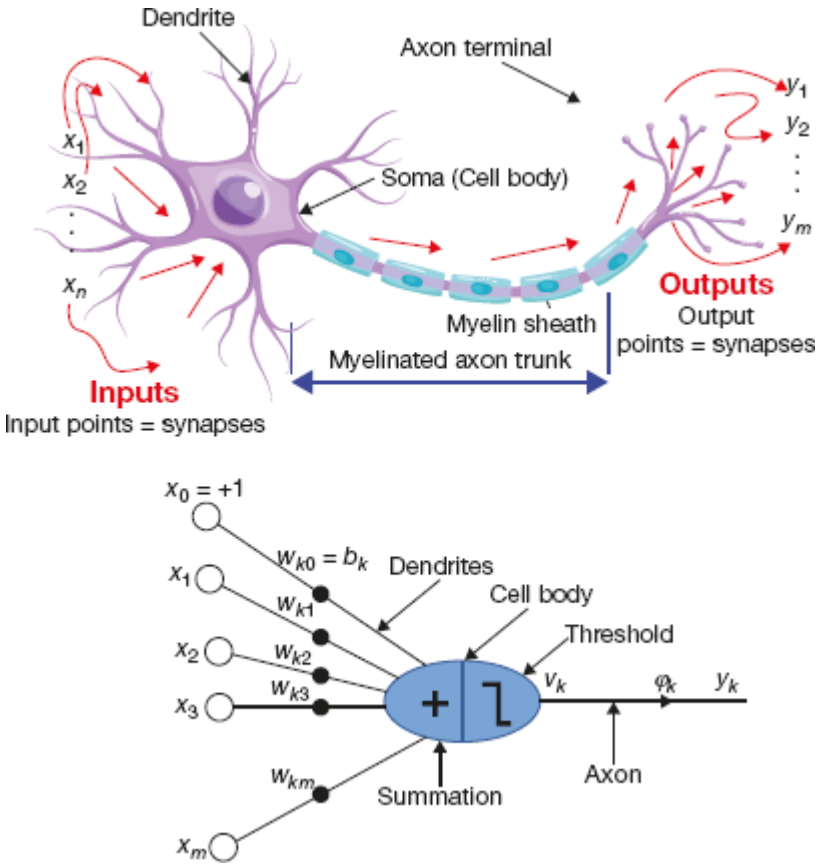


Figure 2.12 Biological neuron with its artificial counterpart

A biological neuron takes inputs through synapses from other neurons. They are aggregated and the summed input may trigger the neuron, in which case the neuron fires and a signal flows through the axon to the dendrites. This simple artificial neural network was also called perceptron. Apparently, these could be used, to simulate a simple decision-making process (Binary Choices).

The below table furnishes the differences between artificial neuron and biological neurons.

Property	Biological Neuron	Artificial Neuron
Composition	Living cells	Computer code
Inputs	Many (dendrites)	Few
Outputs	One (axon)	Many
Signals	Electrical spikes (action potentials)	Continuous values
Speed	Slow (up to 200 Hz)	Fast (millions of Hz)
Energy efficiency	Efficient	Inefficient
Noise level	Noisy	Less noisy
Learning	Yes	Yes
Adaptability	Yes	Yes
Network topology	Complex	Flexible

Let's look at an example here to understand the decision-making process of a neuron. Let's say, you want to decide whether to watch a movie or not. To keep life simple, here our assumption is that you will decide to watch the movie, based on two of your friend's opinions expressed in terms of rating (In a scale of 5). You assign their ratings different weights (well you may rely on one friend's opinion more than the other) and come to a final rating (aggregation). If this combined rating is more than 3, then you will watch the movie (the neuron will fire). This process in illustrated with the help of the below diagram (**Fig. 2.13**).

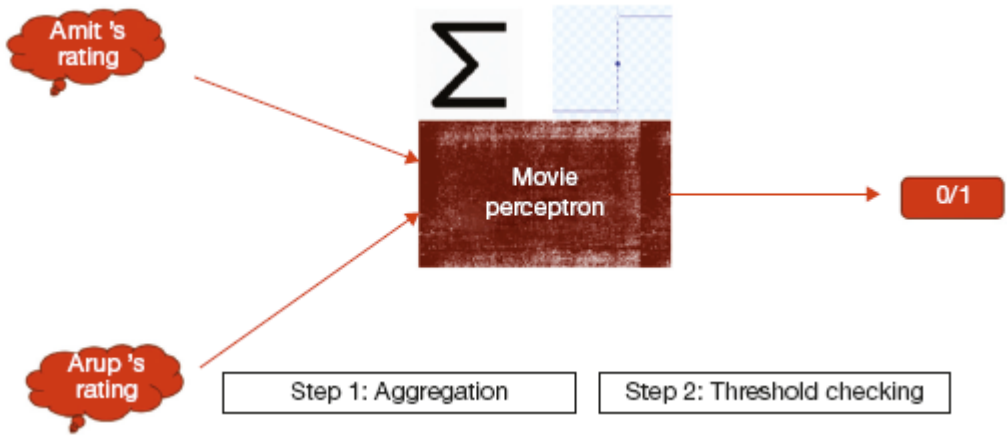


Figure 2.13 Step by step decision making by ANN

Let's say Amit rates the movie 3.5 and Arup rates it 2.5. For some reason, we give a weightage of .6 to Amit and then .4 to Arup. Then the combined rating is 3.5 * .6 + 2.5 * 4 = 2.1 + 1 = 3.1. The threshold as mentioned is 3, hence you decide to watch the movie. Well, without making things two

complicated, let's try to put this mathematically. Let, the rating of Amit is represented by R_1 and the weightage to his rating is represented by W_1.

On a similar note, R_2 and W_2 represent Arup's rating and weightage of rating respectively. The value or threshold which is 3 in this case is presented by W_3. Hence, if the following equation gives a positive value, then, you watch the movie, else you do not.

$$w_1 * x_1 + w_2 * x_2 - w_3 \geq 0$$

Well, if you look at the equality part carefully and concentrate on $w_1 * x_1 + w_2 * x_2 - w_3 = 0$. This actually represents, the equation of a line. The inequality also serves as an activation function.

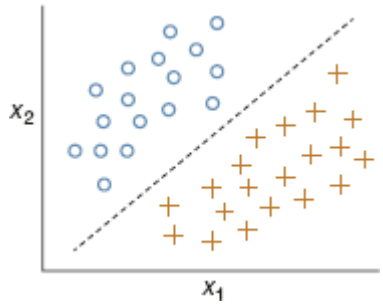


Figure 2.14 Linearly separable classes

If we look at **Fig. 2.14**, the circles and plus symbols represent individual movies. If the ratings of the movies are on the right side of the dotted line, then you will watch the movie, else you won't watch it. The dotted line in this case is called the decision boundary.

The Perceptron algorithm boosted the capability of perceptron further. Given a problem and a few past movies (rating of Amit and Arup for those movies and your choice of watching the movie or otherwise after knowing your friends' rating is available) data is available, then from the data itself, the algorithm can learn the values of w_1, w_2 and w_3. This makes the perceptron a general decision-making tool.

After the initial euphoria was over, one scepticism, that hit the perceptron community was that it will not be suitable for complex problems. They observed, that perceptron can only work when these two classes

(Movies I am going to watch and movies I am not going to watch) are separable by a straight line.

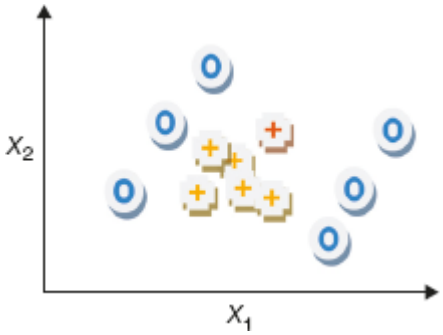


Figure 2.15 Classes not non-linearly separable

For example, in **Fig. 2.15**, the circled movies and the plus sign movies cannot be separated by a line. The problem described may look really trivial, however, the answer remained elusive for quite some time, ushering one of the AI Winters (a period, when there was a lack of optimism, research and funding in anything directly or even remotely linked with AI).

As more research continued, scientists realised while this may not be achievable by a single line, if multiple lines can be employed, a complex decision boundary can be achieved. Such a probable solution is depicted below in the diagram (**Fig 2.16**).

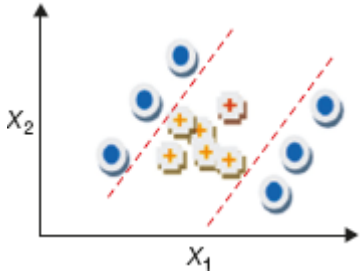


Figure 2.16 Classes not non-linearly separable, separated using two lines

Hence, two perceptrons can be used to draw two lines and then another unit can check whether the movie or observations are falling within the two lines or outside of any of the lines. This led to the theory of the Multi-Layer Perceptron. **Figure 2.17** represents such a setup.

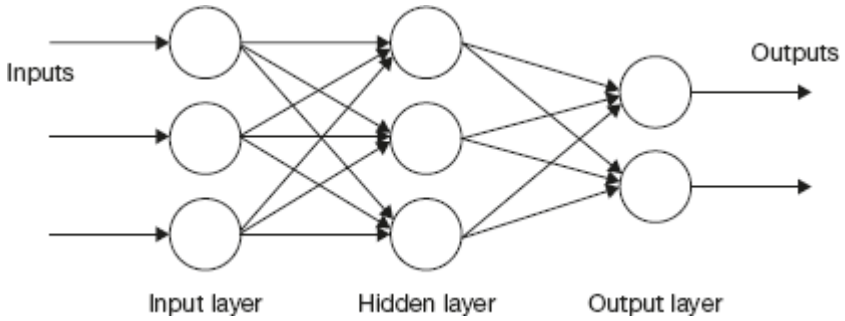


Figure 2.17 Building blocks of a multi-layer perceptorn

Typically, there are three types of layers. The first is the input layer, where it takes inputs. Then, there is the output layer and finally, we have a hidden layer in between. It turns out that, there can be multiple hidden layers and with more hidden layers, the resultant neural network becomes more capable.

There is one more important algorithm, which needs mention.

- In 1989, George Cybenko proposed a universal approximation theorem that showed a feedforward neural network with a single hidden layer and a continuous activation function can approximate any continuous function, given enough hidden units.
- Shortly after Cybenko's work, Kurt Hornik, Maxwell Stinchcombe and Halbert White showed that the same result holds for feedforward neural networks with multiple hidden layers. They also showed that the specific choice of activation function is not important, as long as it is continuous.

The universal approximation theorem is one of the most important results in the field of artificial intelligence. It shows that neural networks are capable of learning any complex pattern in data, as long as the pattern is continuous. Networks, with 2 or more hidden layers are called deep neural networks.

While the concept of multi-layer perceptron came around the 1980s, it took rigorous research for around 25–30 years, to start reaching the potential of these networks. There are three gentlemen namely Geoffrey Hinton, Yoshua Bengio and Yann Lecun, who kept up their research. They

are rightly called the fathers of deep learning. In 2018, they were awarded with the Turing Award (Highest Honour in Computer Science).

One question, that will surely come to your mind is what changed in the course of time, to make deep learning models successful. Below is a summary of the same.

GPU Acceleration

The availability of powerful Graphics Processing Units (GPUs) significantly accelerated the training of deep neural networks, making it feasible to work with large-scale models and datasets. Without too much technical nitty-gritty, GPU may be assumed to be like a supercharged, specialized computer chip inside your computer that's really good at handling tasks related to graphics and calculations. It's like having a separate, powerful brain just for handling images, videos, and complex math problems.

Access to Big Data

The proliferation of digital data and the internet have provided an abundance of labelled and unlabelled datasets, which is essential for training deep learning models effectively. Compared to machine learning models, the deep learning models have more parameters. More the parameters, more is the requirement of data.

Advancements in Algorithms

Breakthroughs in neural network architectures (e.g., convolutional neural networks and transformers) and optimization techniques (e.g., stochastic gradient descent) have greatly improved the performance of deep learning models, enabling them to tackle complex tasks.

- Deep learning has achieved remarkable success in various domains, including computer vision, natural language processing, and speech recognition.
- Convolutional Neural Networks (CNNs) excel in image-related tasks, such as object detection and image classification. CNN is specialized for images. It turns out that an image is formed in our visual cortex in

quite a layered manner. (In 1981, Hubel and Wiesel, got the Nobel prize for explaining the visual processing of the brain).

- Recurrent Neural Networks (RNNs) are well-suited for sequential data and time series analysis. Some examples of sequence data are language, weather data, stock market data, data from different sensors, etc.
- Long Short-Term Memory (LSTM) networks are a specialized form of RNNs designed to capture long-range dependencies in sequences.
- Transformers have revolutionized natural language processing and are the foundation for many state-of-the-art language models. Current sensations like ChatGPT, Bard etc., are based on transformer architecture.
- Transfer learning and pre-trained models have become common techniques for leveraging existing deep learning knowledge. This needs further explanation. Knowledge acquired from one task is very conveniently used by humans in another task. Knowledge of walking can be used in running or climbing stairs. Knowledge of vocabulary can be used for playing scrabble or writing an essay. Knowledge of playing badminton can be used in playing table tennis. Often, it also happens because of your already acquired knowledge of badminton (pre-training), it can be transferred to playing table tennis (With some adjustments, known as fine-tuning). This is possible for deep learning as well. A model trained for one image classification task can be used for another image classification task with minor adjustments. It may be noted that transfer learning was not feasible in machine learning models.

Let’s look at some of the important deep learning architectures.

Model	Description	Use cases
Convolutional Neural Networks (CNNs)	CNNs are a type of deep learning model that is particularly well-suited for image recognition and classification tasks. They work by extracting features from images using a series of convolutional and pooling layers.	CNNs are used in a wide variety of applications, including image classification, object detection, and facial recognition. For example, CNNs are used to power the image search and photo tagging features in Google Photos and Facebook.
Recurrent Neural Networks (RNNs)	RNNs are a type of deep learning model that is well-suited for sequential data, such as text and audio. They work by processing the data one sequence element at a time and maintaining a hidden state that captures the information from the previous elements in the sequence.	RNNs are used in a variety of applications, including natural language processing, machine translation, and speech recognition. For example, RNNs are used to power the Google Translate and Siri features.
Long Short-Term Memory (LSTM)	LSTMs are a type of RNN that is particularly well-suited for tasks that require long-range dependencies, such as machine translation and speech recognition. LSTMs work by using a special type of memory cell to store information from the previous elements in the sequence.	LSTMs are used in a variety of applications, including natural language processing, machine translation, and speech recognition. For example, LSTMs are used to power the Google Translate and Siri features.
Generative Adversarial Networks (GANs)	GANs are a type of deep learning model that consists of two competing networks: a generator network and a discriminator network. The generator network tries to generate realistic data, while the discriminator network tries to distinguish between real and generated data.	GANs are used in a variety of applications, including image generation, text generation, and music generation. For example, GANs are used to power the DeepDream feature in Google Photos, which can generate psychedelic images from ordinary photos.
Transformers	Transformers are a type of deep learning model that is particularly well-suited for natural language processing tasks. Transformers work by using a self-attention mechanism to learn long-range dependencies in the data.	Transformers are used in a variety of natural language processing tasks, including machine translation, text summarization, and question-answering. For example, Transformers are used to power the Google Translate and Bard features.
Autoencoders	Autoencoders are a type of deep learning model that are trained to reconstruct their input data. They consist of two parts: an encoder and a decoder. The encoder learns to compress the input data into a latent representation, and the decoder learns to reconstruct the input data from the latent representation.	Autoencoders are used in a variety of applications, including anomaly detection, dimensionality reduction, and data denoising. For example, autoencoders are used to detect fraudulent credit card transactions and to reduce the dimensionality of images before classification.

2.4.8 Reinforcement Learning

Reinforcement learning is a type of machine learning where an agent learns to behave in an environment by trial and error. The agent receives rewards for taking actions that lead to desired outcomes, and punishments for taking actions that lead to undesired outcomes. Over time, the agent learns to take actions that maximize the expected reward. Reinforcement learning is a powerful technique that can be used to solve

a wide variety of problems, such as playing games, controlling robots and making financial decisions.

Some key concepts of reinforcement learning are as follows:

- **Agent:** The agent/a special software program is the entity that learns to behave in the environment.
- **Environment:** The environment is the world in which the agent acts.
- **State:** The state is the current condition of the environment.
- **Action:** An action is something that the agent can do.
- **Reward:** A reward is a signal that the agent receives for taking an action that leads to a desired outcome.
- **Punishment:** A punishment is a signal that the agent receives for taking an action that leads to an undesired outcome.
- **Policy:** A policy is a rule that the agent uses to select actions.
- **Value function:** A value function is a function that estimates the expected reward for taking an action in a given state.

Below is a simple example of how reinforcement learning can be used to train a robot to walk:

1. The robot starts in a random position.
2. The robot takes an action, such as moving its left leg forward.
3. The environment provides the robot with a reward or penalty, depending on the outcome of the action. If the robot takes a step forward without falling down, it receives a reward. If the robot falls down, it receives a penalty.

The robot repeats steps 2 and 3 until it reaches its goal, or until it runs out of time.

Multi-Armed Bandit

Multi-Armed Bandit is a very fundamental problem as far as reinforcement learning is concerned. A multi-arm slot machine is shown in **Fig. 2.18**.



Figure 2.18 Multi-Armed Bandit

Let's quickly understand what is a single armed bandit. It is a slot machine with a single lever deployed at casinos or often at fairs. The setup is simple, there is a lever, which needs to be pulled after inserting coin or money. When the lever is pulled, some numbers or symbols are displayed. Some of the combinations of symbols are winning combinations, when the player gets some reward in the form of money or tokens. However, in most cases the player loses and the machine ends up robbing money from the players, hence the name bandit. When there are multiple slot machines, this is called as multi-armed bandit problem. Different slot machines will have different probabilities and certainly, the player would want to play in the machine which has the highest winning chances or probability.

Here's a short step-by-step description of how a multi-armed bandit problem works.

Setup

Imagine you are in a casino standing in front of several slot machines (the 'one-armed bandits'). Each slot machine has its own unknown probability of giving you a reward when you pull its lever. Your goal is to maximize your total reward by choosing which slot machines to play and how often to play them.

Exploration vs. Exploitation

In the beginning, you don't know which slot machine is the most rewarding. So, you face a trade-off between exploring (trying out different ma-

chines to learn their winning chances) and exploiting (choosing the machine that seems to be the best based on your current knowledge).

Strategy

You start with a strategy or policy. This strategy dictates how you decide which machine to play at each step.

There are different strategies you can use, such as:

- **Greedy strategy:** Always play the machine that has the highest estimated reward so far.
- **Epsilon-greedy strategy:** Most of the time, play the machine with the highest estimated reward, but occasionally explore other machines randomly (with probability ε). ε is a value between 0 to 1. Value of .1 means 10% exploration, .2 means 20% exploration. These are typical values of ε. However, the value of ε may be changed based on the type of the problem.
- **UCB (Upper Confidence Bound):** Choose machines based on the uncertainty in their estimated rewards. As we pull different levers, we win at different rates. The average reward received for each slot machine can be calculated very easily. UCB also considers the number of trials that were considered, while calculating the average return. If the average return is calculated on the basis of lees no. of trials, then uncertainty is greater. Machines with high uncertainty are explored more.
- **Thompson sampling:** Sample from a probability distribution for each machine and play the machine with the highest sampled value. Here, the experiment starts with different prior assumptions about the reward/winning chances associated with the machines.

Repeated Actions

We start pulling the levers of the chosen machines according to our strategy. After each pull, we observe whether we win a reward or not.

Learning

As we play more, we update our estimates of each machine's reward probabilities based on the outcomes we have observed. The goal is to improve the accuracy of your estimates to make better decisions.

Iteration

We repeat steps 3–5 for a certain number of rounds or until we achieve a desired level of reward.

Adaptation

Over time, our strategy adapts as we learn more about the machines' reward probabilities, shifting the balance between exploration and exploitation.

The multi-armed bandit problem is a fundamental concept in reinforcement learning, decision-making under uncertainty, and optimization. It has applications in various fields, including online advertising, clinical trials, and recommendation systems, where you need to make choices in an uncertain environment.

Q-Learning

Q-Learning is another important paradigm in reinforcement learning. Multiple arm bandit deals with uncertain environments, but there are some tasks which are continuous in nature. In Q-learning, an agent learns a policy to maximize cumulative rewards by interacting with an environment with multiple states, actions, and rewards. The agent interacts with an environment characterized by states and actions. It learns a Q-value for each state-action pair, allowing it to make sequential decisions based on its current state and knowledge of the Q-values for available actions. The learning process is typically more involved, as the agent must explore, learn, and exploit to optimize its decisions over time. Q-learning is well-suited for problems that involve sequential decision-making, such as game playing, robotics, and control systems. It can handle more complex environments with state transitions. Markov Decision Process (MDP) is linked with Q Learning.

A Markov Decision Process (MDP) is a mathematical framework used to model decision-making problems in a stochastic environment. Here's a brief explanation of an MDP:

- **States (S):** An MDP represents a system or environment that can be in various states. These states could be discrete (e.g., grid cells in a game) or continuous (e.g., real numbers in a control system).
- **Actions (A):** At each state, an agent can take a set of possible actions. These actions are the choices available to the agent to influence the environment.
- **Transitions (T):** When the agent takes an action in a particular state, the system transitions to a new state. The transition function, T, describes the probabilities of moving from one state to another based on the chosen action.
- **Rewards (R):** At each state, the agent receives a numerical reward, which represents the immediate benefit or cost of being in that state. The reward function, R, defines these rewards.
- **Policy (π):** A policy is a strategy or plan that guides the agent's decision-making. It specifies which action to take in each state.
- **Goal:** The objective in an MDP is typically to find an optimal policy that maximizes the cumulative reward over time.

2.5 ROBOTICS

Robotics is the interdisciplinary field of engineering, science, and technology that deals with the design, construction, operation, and application of robots. Robots are machines that can sense their environment and take actions to achieve specific goals. They can be used for a variety of tasks, such as manufacturing, healthcare, and customer service. The field of robotics is constantly evolving, as new technologies are developed to make robots more intelligent, autonomous, and adaptable.

2.5.1 Component of a Robot

Let us quickly look at the basic components of a robot.

- **Frame:** The frame provides the robot with its structure and shape. It is usually made of metal or plastic.

- **Actuators:** Actuators are the components that move the robot. They can be motors, pneumatic cylinders, or hydraulic cylinders.
- **Sensors:** Sensors allow the robot to interact with its environment. They can detect things like light, heat, touch, and sound. Common types of sensors include:
  - Proximity Sensors: Detect the presence of nearby objects or obstacles.
  - Light Sensors: Measure ambient light levels.
  - Temperature Sensors: Measure the temperature of the environment.
  - IR (Infrared) Sensors: Detect infrared radiation and can be used for distance measurement.
- **Controller:** The controller is the brain of the robot. The control system of a robot includes the hardware and software that process sensor data and make decisions on how the robot should respond. It can range from simple microcontrollers to complex computing systems. The control system determines how the actuators respond to changes in the environment as detected by the sensors.
- **Power supply:** The power supply provides the robot with the electricity it needs to operate.
- **Communication Interfaces:** Robots may need to communicate with external devices or systems. Communication interfaces such as Wi-Fi, Bluetooth, USB ports, and serial connections enable data exchange between the robot and other devices.
- **End Effectors:** If the robot is designed to interact with objects or perform tasks, it may have specialized end effectors or grippers. These are devices attached to the robot's manipulator that allow it to grasp, manipulate, or interact with objects.

2.5.2 Types of Robots

The robots may be classified based on where they are employed.

- **Industrial robots:** These robots are used in factories to perform repetitive tasks, such as welding, painting, and assembly. They are typically large and strong, and they can be programmed to perform a variety of tasks.

- **Service robots:** These types of robots are used to perform tasks that are helpful to humans, such as cleaning, delivering food, and providing customer service. They are typically smaller and more agile than industrial robots, and they can be programmed to interact with humans in a safe and efficient way.
- **Medical robots:** These robots are used in surgery and other medical procedures. They can be used to perform delicate tasks with greater precision than human surgeons.
- **Military robots:** These robots are used in warfare to perform activities that are considered dangerous or difficult for humans, such as bomb disposal and surveillance.
- **Research robots:** These robots are used to conduct research in a variety of fields, such as robotics, artificial intelligence and materials science.
- **Educational robots:** These robots are used to teach people about robotics and STEM subjects. They are typically small and easy to use, and they can be programmed to perform a variety of tasks.
- **Entertainment robots:** These robots are used for entertainment purposes, such as dancing, performing and companionship. They are typically designed to be interactive and engaging.
- **Manufacturing:** Robots are used in manufacturing to perform tasks such as welding, painting, and assembly. They can help to improve efficiency and accuracy, and they can also reduce the risk of injuries to workers.
- **Healthcare:** Robots are used in healthcare to perform tasks such as surgery, rehabilitation and patient care. They can help to improve the quality of care and reduce the cost of healthcare.
- **Customer service:** Robots are used in customer service to answer questions, provide support, and complete transactions. They can help to improve customer satisfaction and reduce the workload of human employees.
- **Military:** Robots are used in the military to perform tasks such as surveillance, bomb disposal and combat. They can help to protect soldiers and civilians, and they can also increase the effectiveness of military operations.
- **Space exploration:** Robots are used to explore space and perform tasks that are too dangerous or difficult for humans. They can help to

gather data, conduct experiments and build structures on other planets.

2.5.3 Sophia

Sophia is a humanoid robot (**Fig. 2.19**) developed by Hong Kong-based Hanson Robotics. It gained international attention due to its lifelike appearance and advanced artificial intelligence capabilities. Sophia was designed to visually resemble a human and can simulate facial expressions and hold conversations with people.



Figure 2.19 Sophia is the first robot citizen

Key features of Sophia include:

- **Appearance:** Sophia is designed to mimic human features, with a face that can move its eyes, lips, and eyebrows, creating a range of facial expressions.
- **AI Interaction:** Sophia's artificial intelligence system enables it to engage in conversations with humans. It uses natural language processing to understand and respond to questions and statements.
- **Voice Recognition:** Sophia has the ability to recognize and process speech, allowing it to respond to verbal commands and questions.

- **Learning Abilities:** While not a fully autonomous learning machine, Sophia can learn and adapt to some extent through interactions. It can remember individual users and previous conversations.
- **Public Appearances:** Sophia has been showcased at various international events, conferences, and media appearances. It has even been granted citizenship in some countries, although this symbolic gesture has generated ethical discussions about robot rights and responsibilities.
- **Expression of Emotion:** Sophia can display a limited range of emotional expressions using its facial features, allowing it to convey happiness, surprise, sadness and other emotions.



DO YOU KNOW?

ImageNet is a large-scale image database widely recognized for its role in advancing computer vision and machine learning. Founded by Fei-Fei Li in 2007, ImageNet contains millions of labelled images in various categories. It gained fame through the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), an annual competition that spurred the development of deep learning algorithms, particularly convolutional neural networks (CNNs). In 2012, the breakthrough came when a CNN-based approach achieved a significant drop in classification error. This marked a pivotal moment in the history of AI, paving the way for improved object recognition, image classification, and image analysis. ImageNet remains a valuable resource for researchers and has significantly contributed to the progress of computer vision technology.

2.6 KNOWLEDGE ENGINEERING

Knowledge engineering is a discipline within artificial intelligence (AI) and computer science that focuses on designing, building, and maintaining knowledge-based systems. The goal of knowledge engineering is to capture, represent, organize and utilize human expertise and domain-specific knowledge in a format that can be processed and manipulated by computers. This knowledge is then used to create intelligent systems that

can reason, make decisions, and solve complex problems in specific domains.

2.6.1 Steps in Knowledge Engineering

1. **Knowledge Acquisition:** This involves gathering knowledge from domain experts and converting it into a format that can be understood by computers. Different techniques, such as interviews, surveys and analysis of existing documents, are used to capture the expertise of human experts.
2. **Knowledge Representation:** Once knowledge is acquired, it needs to be organized and structured in a way that can be easily interpreted by machines. Various formalisms, such as ontologies, semantic networks, rules, frames, and more, are used to represent knowledge in a format suitable for AI systems.
3. **Knowledge Integration:** Often, knowledge engineering involves integrating various sources of knowledge to create a comprehensive representation. This can include combining structured data, unstructured text and expert rules to build a cohesive knowledge base.
4. **Inference and Reasoning:** Knowledge engineering involves developing mechanisms and algorithms for drawing conclusions, making inferences, and reasoning with encoded knowledge. This allows the AI system to answer questions, solve problems and simulate human-like decision-making.
5. **Maintenance and Refinement:** As domains evolve, knowledge-based systems need to be updated and refined to remain accurate and relevant. Knowledge engineers are responsible for maintaining and improving the knowledge bases over time.
6. **Verification and Validation:** Ensuring the correctness and reliability of the knowledge base is crucial. Verification involves checking that the knowledge has been accurately captured and represented, while validation involves confirming that the system's outputs align with expected outcomes.
7. **Deployment and Interaction:** Knowledge engineering includes integrating the knowledge-based system into real-world applications and enabling interaction between the system and users. This often in-

volves creating user interfaces that allow users to query the system and receive meaningful responses.

Among the above steps, while some are engineering steps, some steps need more discussion like Knowledge Representation.

2.6.2 Knowledge Representation

There are several different knowledge representation schemes used in artificial intelligence (AI) to capture and represent information and knowledge in a format that computers can understand and manipulate. Each scheme has its own strengths and weaknesses, making them suitable for different types of problems and domains. Here are some common knowledge representation schemes.

Semantic Networks

Semantic networks (**Fig. 2.20**) represent knowledge using nodes (concepts) and links (relationships) between them. This graphical representation is particularly useful for illustrating relationships and hierarchies within a domain.

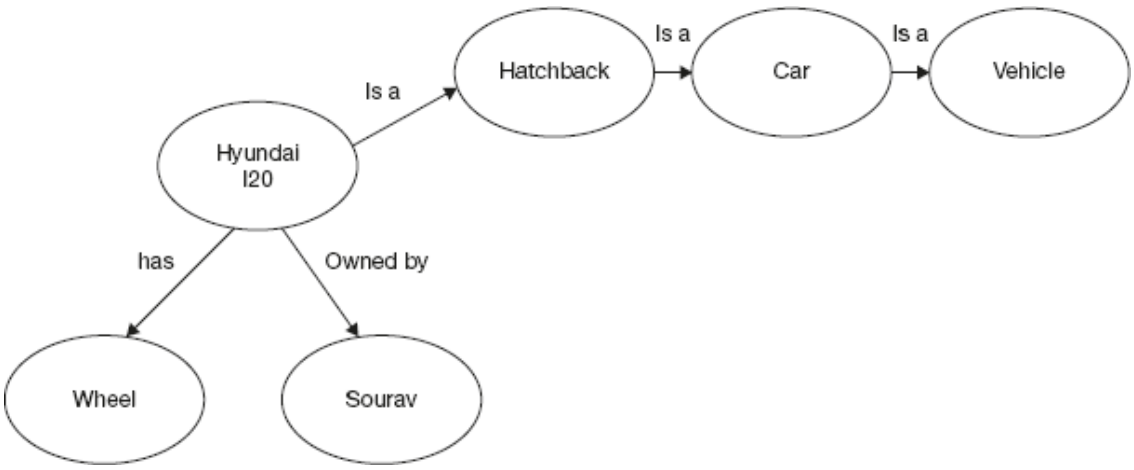


Figure 2.20 A sample semantic network

In the above diagram, a simple concept is presented. Hyundai i20 is a hatchback car, which is also car and also a vehicle. It has wheels and is owned by Sourav. While Semantic Network is a graphical representation with associated benefits, this lacks formalism. There is no clear guideline on types of relationships.

Frames

Frames organize knowledge into structured units called frames, which contain attributes, slots and values that describe an object or concept. Frames are useful for representing structured information and capturing inheritance relationships. If we take the same example, the car will be called a frame and different attributes of the frame like the make, model, year and colour will be called slots.

1. **Make:** Hyundai
2. **Model:** i20
3. **Year:** 2022
4. **Colour:** Red

For representing the concept of hierarchy, concepts like super frames, subframes may be used. For example, here car can be a superframe.

First-Order Logic (FOL)

FOL extends propositional logic by introducing variables, quantifiers and predicates. It provides a formal and expressive representation for relationships, rules and facts. Let's look at the components

- **Variables:** Variables are used to represent objects in the world. They are denoted by lower case letters, such as x, y, and z.
- **Constants:** Constants are used to represent specific objects in the world. They are denoted by upper case letters, such as A, B, and C.
- **Predicates:** Predicates are used to represent relationships between objects. They are denoted by capital letters, such as P, Q, and R.
- **Connectives:** Connectives are used to combine statements together. They include logical operators such as ∧ (and), ∨ (or), → (implies), and ↔ (iff).

FOL statements are constructed by combining these symbols using the following rules:

- A predicate can be applied to a variable or a constant to form a statement. For example, the statement P(x) means that the object denoted

by x has the property denoted by P.
- Statements can be combined using the connectives. For example, the statement P(x) ∧ Q(y) means that the object denoted by x has the property denoted by P and the object denoted by y has the property denoted by Q.

Examples of Objects and Predicates:

- **Predicate:** L (x, y) meaning 'x Likes to eat y'
- **Statement:** 'Soham loves Pizza.'
- **FOL Representation:** L (Soham, Pizza)

Statement with Quantifiers:

- **Universal Quantification:** 'All men are Mortal.'
  - **FOL Representation:** ∀x (Man(x) → Mortal(x))
  - Where ∀x means 'for all x,' Man(x) represents 'x is a Man,' and Mortal(x) represents 'x is a Mortal.'
  - The arrow indicates implies
- **Existential Quantification:** 'There exists a red car.'
  - **FOL Representation:** ∃x (Car(x) ∧ Red(x))
  - Where ∃x means 'there exists an x,' Car(x) represents 'x is a car,' and Red(x) represents 'x is red.'

There are many other notations, like Sunny(x), which stands for negation. Essentially, this means x, which represents that the day x is not sunny.

Rules/Production Rules

Rule-based systems use a set of if-then rules to represent knowledge. Production rules, also known as production systems, are a fundamental concept in the field of artificial intelligence (AI) and expert systems. They are used for representing knowledge and making decisions based on a set of condition-action pairs. Production rules consist of two main components:

Conditions

These are logical statements or tests that describe the circumstances or criteria that must be met for the rule to be applied. Conditions are typically expressed in terms of facts, data, or variables. They define when a particular rule should be triggered.

Actions

These are statements or instructions that specify what should be done when the conditions associated with a rule are satisfied. Actions can include making conclusions, updating facts, executing procedures, or triggering other rules.

Production rules are often written in the following format:

IF (Conditions)

THEN (Actions)

Here are a few examples of production rules:

Medical Diagnosis:

IF patient has a fever AND patient has a sore throat

THEN diagnose the patient with the flu AND prescribe antiviral medication

Inventory Management:

IF inventory level is below a certain threshold

THEN reorder the product in a specified quantity

Production rules are a powerful knowledge representation tool because they allow AI systems to encode domain-specific knowledge and expertise in a structured and understandable manner. Expert systems, which are AI systems designed to solve specific problems or make decisions in spe-

cialized domains, often use production rules to model the decision-making process. When applied sequentially or in parallel, production rules enable expert systems to reason, infer and provide solutions based on the available data and knowledge.

Few other important topics as far as knowledge engineering is concerned is Fuzzy logic.

2.6.3 Fuzzy Sets and Logic

A fuzzy set is a concept in mathematics and set theory that extends the idea of a crisp set. To understand a fuzzy set, it's helpful to first define a crisp set.

A crisp set is a well-defined set-in traditional set theory. It contains elements that either completely belong to the set or completely do not belong to the set. For example, in a crisp set of 'Even Numbers,' 2, 4, and 6 are fully part of the set, while 3, 5, and 7 are completely outside the set.

A fuzzy set is a more flexible and nuanced concept. In contrast to crisp sets, fuzzy sets allow elements to have degrees of membership. This means that elements can partially belong to the set to varying degrees. Instead of just 'in' or 'out' of the set, you can say an element is, member of a set with membership value of 0.7, indicating a high degree of membership.

Example

Let's consider a fuzzy set of 'Tall People' based on height. In a crisp set, you would say someone is either 'in' the set (tall) or 'out' of the set (not tall). However, in a fuzzy set, you can express degrees of tallness. For instance, a person who is 6 feet tall might have a membership degree of 0.8, indicating a high degree of being 'tall,' while someone who is 5 feet tall might have a membership degree of 0.3, showing a lower degree of 'tallness.' Fuzzy logic is a type of logic that uses fuzzy sets to represent knowledge and to make decisions. Fuzzy logic is based on the idea that there are many different degrees of truth, rather than just true and false.

Fuzzy logic is used for diverse applications, such as:

1. **Robotics:** Fuzzy logic is used to control robots in complex and uncertain environments.
2. **Machine learning:** Fuzzy logic is particularly useful for machine learning algorithms which have to deal with data that is noisy or incomplete.
3. **Natural language processing:** Fuzzy logic is used to construct natural language processing systems that can understand and generate human language in a more natural way.

Here is an example of how fuzzy logic can be used to make a decision:

Suppose you are designing a robot that needs to be able to navigate in a crowded environment. You could use fuzzy logic to represent the robot's knowledge about the environment, such as the distance to other objects and the speed of other objects. You could then use fuzzy logic to make decisions about how the robot should move. For example, the robot could use fuzzy logic to decide whether to turn left, turn right, or go straight.

Fuzzy logic is a very effective tool that can be used to model and reason about uncertainty. It is used in a variety of applications to make decisions in complex and uncertain environments.



DO YOU KNOW

Fuzzy String matching is a popular concept, where when a string is searched, instead of exact match (crisp), similarity score is calculated and similar strings are returned. This is called as partial matching. Suppose you are searching for the word 'cat' on a website. The website's search engine uses fuzzy string matching to find all of the pages on the website that contain words that are similar to 'cat', such as 'cats', 'kitten', and 'kitty cat'.

Rough set theory is a mathematical framework used to analyse data when you have incomplete or uncertain information. It helps you make deci-

sions and classify data into different categories, even when you're not entirely sure.
Let's look at another important concept named as Rough Set.

Example 1: Imagine you have a collection of fruits, and you want to classify them as either 'Apples' or 'Oranges' based on their characteristics. You have information about their colour, taste and shape.

1. **Definitely Apples**: These are the fruits you're quite certain are apples based on the information you have. For instance, if a fruit is red, sweet, and round, you can confidently say it's an apple.
2. **Possibly Apples**: These are the fruits that you're not entirely sure about. They might be apples, or they might be oranges. For instance, a fruit could be red and round (like an apple), but you're not sure about its taste. It might be sweet (like an apple) or sour (like an orange).

In rough set theory, 'Definitely Apples' represents the lower approximation, and 'Possibly Apples' represents the upper approximation. The upper approximation includes fruits that might belong to the 'Definitely Apples' group, but you can't be completely certain.

Example 2: Now, let's say you have a dataset of cars, and you want to determine if they are 'Fuel-Efficient' or 'Not Fuel-Efficient.' You have information about their fuel efficiency (miles per gallon), engine size, and weight.

1. **Definitely Fuel-Efficient Cars**: These are the cars you're quite certain are fuel-efficient based on the information you have. For example, if a car has high fuel efficiency, a small engine, and is lightweight, you can confidently classify it as fuel-efficient.
2. **Possibly Fuel-Efficient Cars**: These are the cars that you're not entirely sure about. They might be fuel-efficient, or they might not be. For instance, a car might have high fuel efficiency (like a fuel-efficient car) but a large engine (which is not typical for fuel-efficient cars). It falls into the 'Possibly Fuel-Efficient' category because you can't be completely sure.

Rough set theory helps you work with data like this, where you have uncertainty, and it allows you to make decisions and classifications even when some information is incomplete or ambiguous.

The following table encloses the difference between fuzzy and rough sets:

Characteristic	Fuzzy set	Rough set
Definition	A set where the elements have degrees of membership between 0 and 1.	A set where there are some members that belong to the set with certainty and some members that may or may not belong to the set.
Uncertainty	Uses degrees of membership to represent the uncertainty in whether or not an element belongs to a set.	Uses crisp sets to represent the uncertainty of whether or not an element belongs to a set.
Knowledge representation	Typically used to represent knowledge about the world.	Typically used to extract knowledge from data.
Example	The fuzzy set of 'tall people'.	The rough set of 'customers who are likely to make a purchase'.

2.7 SUMMARY

- The ultimate goal of AI should be creating an artificial man/robot.
- The subsystems for the same will be computer vision, natural language processing, robotics, machine learning and knowledge representation.
- The objective of computer vision is to enable computers to understand a natural scene like a human being can. Some important areas of computer vision, are object recognition, face recognition, medical imaging, etc.
- Similarly, the goal of NLP is to develop the ability of computers to understand language and generate responses in natural language. Some of the very critical tasks are text summarization, question answering, sentiment analysis etc.
- Machine learning concerns about the efficient algorithm for learning. The basic types are supervised learning where you have past examples and unsupervised learning which does not have past examples. Some important algorithms are k-means, k-NN, decision tree and logistic regression.
- One other sub-field of machine learning is called deep learning, whose basic unit is an artificial neuron, this is mimicked after a biological neuron.

- Deep learning debuted with limited success but then picked up as more data became available, there is significant algorithmic improvement and GPU.
- Reinforcement learning is a subfield, which is best suited for trial-and-error environment. The best policy is determined as the agent interacts with the environment through actions (For desired actions, agents receive reward and for not-so-desired actions the agent receives penalty), some of the important paradigms of reinforcement learning are multi-armed bandit, Q-Learning, etc.
- Some of the important components of robots are sensors, actuators, frames, communication interface, power supply, etc.
- Some of the important application domains where robots are being used are education, military, health, industrial automation, etc.
- Knowledge representation/engineering is another important component. For representation techniques like Semantic Net, Frames can be used. Predicate and first-order logic can be used for reasoning. For dealing with uncertainty concepts like fuzzy set and logic and rough set can be employed.

2.8 PRACTICE EXERCISES

2.8.1 Subjective Questions (10 Marks)

1. What is the fundamental difference between intelligent thinking (cognitive intelligence) and intelligent action (behavioural intelligence) in the context of AI? Provide examples for each.

2. What are some of the most important capabilities that AI systems must have that function at the human or superhuman level? Give a thorough explanation of each capacity.

3. How do the different subfields of AI intersect and collaborate with each other to advance the field of artificial intelligence? Give examples of how advancements in one subfield can benefit others

4. Explain the concept of object recognition in computer vision. Provide examples of real-world applications where object recognition is used.

5. Describe the difference between face detection and face recognition in computer vision. Provide real-world examples of each of these techniques in detail.

6. Discuss the importance of scene understanding in computer vision and provide examples of how it is used in applications like self-driving cars and augmented reality.

7. How is computer vision technology being utilized in the field of medical imaging? What benefits and challenges does it bring to the healthcare industry?

8. Explore the potential applications and challenges of computer vision in the development of the metaverse and its impact on social interactions and digital experiences.

9. Explain the importance of text pre-processing in Natural Language Processing (NLP). Provide examples of common text pre-processing techniques.

10. Explain the key differences between biological neurons and artificial neurons in the context of deep learning. How do artificial neurons simulate the decision-making process of biological neurons?

11. What is the role of the universal approximation theorem in deep learning, and how does it relate to neural network architecture? Provide an example to illustrate its significance.

12. Describe the main steps in knowledge engineering and explain the importance of each step in the development of knowledge-based systems. Provide an example to illustrate the process.

13. Discuss the various knowledge representation schemes used in artificial intelligence, including Semantic Networks, Frames, First-Order Logic (FOL), and Production Rules. Compare and contrast their strengths and weaknesses, and provide a real-world example where each representation scheme might be suitable.

14. Explain the concepts of Fuzzy Sets and Logic in the context of knowledge engineering. How are fuzzy sets used to represent and process uncertain or vague information? Provide a practical application of fuzzy logic in a specific domain.

2.8.2 Subjective Questions (5 Marks)

1. What a multi-armed bandit problem is different from Q learning.

2. Explain the fundamental concepts of the multi-armed bandit problem in reinforcement learning.

3. Describe the concept of transfer learning in deep learning. How does it enable knowledge transfer between different tasks, and why is it more feasible in deep learning compared to traditional machine learning

4. Describe the concept of dimensionality reduction in machine learning. Why is it important, and what techniques are commonly used for this purpose?

5. Provide an overview of anomaly detection in machine learning. How can it be applied to real-world scenarios, and what are the key use cases?

6. Discuss the challenges and applications of Named Entity Recognition (NER) in NLP. Provide examples of how NER can be applied to real-world scenarios.

7. Explain the fundamental difference between supervised learning and unsupervised learning in machine learning. Provide examples for each.

8. What are the challenges associated with solving the “Long Tail Problem” in recommender systems? How can these challenges be addressed?

9. Explain how object recognition is used in Medical Imaging.

2.8.3 Short Answer-type Questions (2 Marks)

1. Describe the difference between face detection and face recognition in computer vision.

2. Enlist two key differences between rough set and crisp set.

3. Describe the difference between extractive and abstractive text summarization methods in NLP.

4. Differentiate between Intelligent Thinking with Intelligent Action.

5. Explain one important application of Computer Vision adopted in India.

6. Explain briefly how rule-based machine translation works.

7. What is collaborative filtering in context to recommender systems?

8. What do you understand by cold start problem?

9. What is the role of threshold in a neural network?

10. What is the specialization of CNN?

11. What is value function in Reinforcement Learning?

12. What do you understand by exploitation exploration trade off?

13. How Thomson Sampling differs from UCB?

14. What are the common sensors used in Robotics?

15. What are slots in context to frames?

2.8.4 Multiple-choice Questions (1 Mark)

1. What is the key difference between intelligent thinking and intelligent action in AI?

(a) Intelligent thinking involves problem-solving, while intelligent action pertains to understanding natural language.

(b) Intelligent thinking refers to interacting with the physical world, while intelligent action involves reasoning.

(c) Intelligent thinking relates to simulating human thought processes, and intelligent action deals with manipulating the physical world.

(d) There is no difference; both terms are used interchangeably in AI.

2. Which subfield of AI is responsible for processing visual cues and understanding images and videos?

(a) Machine Learning

(b) Knowledge Engineering

(c) Natural Language Processing

(d) Computer Vision

3. When AI systems are capable of self-learning and deriving rules from examples, which subfield does this represent?

(a) Natural Language Processing

(b) Knowledge Engineering

(c) Machine Learning

(d) Robotics

4. What does the subfield of robotics primarily focus on in AI?

(a) Developing algorithms for data analysis

(b) Interaction between computers and human languages

(c) Creating intelligent robots that can interact with the physical world

(d) Knowledge representation and logical inference

5. What is the primary goal of object recognition in computer vision?

(a) To capture images of objects

(b) To identify the presence of a human face

(c) To gain an understanding of what is in an image

(d) To create a realistic virtual environment

6. What is the key difference between face detection and face recognition in computer vision?

(a) Face detection identifies the identity of a person based on their face, while face recognition focuses on the presence of a human face in an image or video.

(b) Face detection is more advanced and specific, while face recognition is more basic and general.

(c) Face detection is used in self-driving cars, while face recognition is used in augmented reality.

(d) Face detection is used for identifying objects in the environment, while face recognition is used for medical imaging.

7. Which emerging concept is closely related to augmented reality and virtual reality and involves social interactions in a digital universe?

(a) Self-driving cars

(b) Object recognition

(c) The metaverse

(d) Scene understanding

8. Which of the following is not a common text pre-processing technique in NLP?

(a) Lowercasing

(b) Tokenization

(c) Image segmentation

(d) Punctuation removal

9. What is the primary goal of text classification in NLP?

(a) Translate text from one language to another

(b) Categorize text into predefined labels

(c) Generate summaries of text

(d) Identify and extract named entities from the text

10. Which machine learning task is focused on predicting continuous numerical values based on input data?

(a) Classification

(b) Clustering

(c) Regression

(d) Dimensionality reduction

11. Anomaly detection in machine learning is used for:

(a) Categorizing data into predefined classes.

(b) Predicting continuous values.

(c) Identifying unusual or abnormal data points.

(d) Reducing the dimensionality of data.

12. The universal approximation theorem in deep learning states that:

(a) A single-layer neural network can approximate any function.

(b) A neural network with multiple hidden layers can approximate any continuous function.

(c) The choice of activation function is irrelevant in neural networks.

(d) Neural networks can only approximate simple linear functions.

13. Transfer learning is more feasible in deep learning than in traditional machine learning because:

(a) Deep learning models have fewer parameters.

(b) Deep learning models require larger amounts of labelled data.

(c) Deep learning models have more complex architectures.

(d) Deep learning models can leverage existing knowledge for new tasks with little adjustments.

14. Which of the following strategies in the multi-armed bandit problem involves playing the machine with the highest estimated reward most of the time but occasionally exploring other machines?

(a) Greedy strategy

(b) Epsilon-greedy strategy

(c) UCB (Upper Confidence Bound) strategy

(d) Thompson sampling strategy

15. What is the primary goal of reinforcement learning?

(a) To classify data into predefined categories

(b) To minimize the difference between predicted values and actual outcomes

(c) To learn how to behave in an environment by trial and error to maximize expected rewards

(d) To generate realistic data using generative adversarial networks

16. What is the primary goal of knowledge engineering?

(a) To design computer hardware

(b) To develop algorithms for natural language processing

(c) To capture, represent, and utilize human expertise and domain-specific knowledge for intelligent decision-making

(d) To create artificial neural networks

2.9 ANSWER KEYS

2.9.1 Multiple-choice Questions

1. (c)

2. (d)

3. (c)

4. (c)

5. (c)

6. (a)

7. (c)

8. (c)

9. (b)

10. (c)

11. (c)

12. (b)

13. (d)

14. (b)

15. (c)

16. (c)